

Bike Rental Demand Forecasting

From Data Preparation to Model Comparison and Critical Reflection

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Introduction to Machine Learning
Group Project - Final Deliverable

Abstract—Accurately forecasting bike rental demand is essential for optimizing modern bike-sharing operations. This report presents a comprehensive machine learning workflow to predict daily rentals using Capital Bikeshare usage statistics and historical meteorological metrics. We formulate the forecasting task as a regression problem and establish a structured data processing pipeline focused on high-traffic stations to ensure reliable signal extraction. To simulate real-world forecasting conditions, we evaluate eight different regression models—ranging from linear baselines to advanced ensemble techniques—under a strict chronological train-validation-test split. A detailed performance comparison using multiple error metrics, including RMSE, MAE, and R^2 , reveals that tree-based ensemble methods significantly outperform linear models. Gradient Boosting proved most effective during validation, while Random Forest achieved the best generalization on the held-out test set. Finally, we critically reflect on these results, concluding that while weather and calendar features provide strong predictive power, future iterations must address uncaptured temporal dependencies to further enhance operational forecasting.

I. INTRODUCTION

Accurately predicting bike rental demand is a crucial operational challenge for bike-sharing systems. The goal of this project is to forecast this demand using machine learning in a way that is technically sound, reproducible, and easy to justify.

To achieve this, we frame the task as a regression problem, utilizing daily data from the Capital Bikeshare system combined with historical weather metrics. This report combines two core phases of the project into one coherent document:

First, it outlines the complete data preparation and feature engineering process. Explaining the final reduced dataset is essential, because the modelling results only make sense if the underlying data representation is clear and robust.

Second, it presents the modelling pipeline and the final evaluation. We train and compare eight different regression models under a strict chronological train-validation-test split. The report provides a detailed performance analysis using error metrics such as RMSE, MAE, and R^2 . By summarizing these results in comparative tables and visual diagnostics, we evaluate which models perform best, interpret their behavior, and critically discuss the current limitations of our forecasting system.

II. DATASET AND FEATURE ENGINEERING

A. Data Origin and Structure

The dataset used in this project is based on the Capital Bikeshare system from the Washington D.C. area. It contains historical bike rental and weather data from May 2020 to August 2024. In our final analytical table, each row represents exactly one day at one specific station. This daily aggregation aligns the demand outcome with the available weather data.

B. The Target Variable

The main goal of our machine learning models is to predict the daily bike demand. Therefore, the task was formulated as a clean regression problem with a single numeric target variable.

TABLE I
TARGET VARIABLE

Column Name	Description
total_rentals	The absolute number of bikes rented at a specific station on a given day.

C. Temporal and Weather Features

To make accurate predictions, the models need to understand seasonal patterns and weather conditions, as both strongly influence the decision to ride a bike. We replaced string-based dates with numerical time representations to make them trainable for our models.

TABLE II
TEMPORAL FEATURES

Column Name	Description
year	The year of the observation.
month	The month (1 to 12) to capture seasonal changes.
weekday	The day of the week (0 to 6) to separate commuting days from weekends.
time_idx	A continuous chronological index. It helps the model to recognize long-term trends over the years.

The dataset also contains several daily aggregated weather metrics. Before finalizing the features, we analyzed correlations to remove redundant weather variables (like `feelslike` temperature).

TABLE III
WEATHER FEATURES

Column Name	Description
tempmax	The maximum temperature of the day (in Celsius).
humidity	The average relative humidity (in percent).
precip	The absolute amount of precipitation (in mm).
precipcover	The percentage of the day with precipitation.
cloudcover	The average cloud cover (in percent).
windspeed	The measured wind speed.
visibility	The average visibility distance.
sealevelpressure	The atmospheric pressure at sea level.
uvindex	The strength of the UV radiation.
sunset_minutes	The time of sunset, expressed in minutes from midnight. This acts as an indicator for daylight duration.
snow	The amount of fresh snow (in mm).
snowdepth	The total depth of snow on the ground (in mm).

D. Station Filtering and One-Hot Encoding

Instead of using all available stations, we decided to focus on the 20 most popular start stations. This was a methodological decision to reduce noise from rare stations and to provide a stronger, more stable signal for the models. It also reduced the data size and allowed for faster experimentation.

In earlier stages, the station identity was stored as a single numeric identifier (`start_station_id`). However, this creates a major problem: machine learning models might wrongly interpret this ID as a continuous mathematical value, assuming that a meaningful distance exists between the numbers.

To fix this weakness, we removed the ID column and applied one-hot encoding. We created 20 separate binary columns, one for each of the top 20 stations.

TODO: Reduced Data and why implementieren! (Tabellen / Grafiken verwenden)

hier kommt die modeling pipeline mit allen models

hier kommen die Auswertung zu den Modellen (Fehlern, Problemen, Anpassungen) + Grafiken/Tabellen

III. PRELIMINARY ETHICAL CONSIDERATIONS

Predicting bike rental demand is generally a low-risk machine learning task. However, we still need to consider some ethical and social effects.

First, we decided to use only the 20 most used start stations. This was necessary to build a stable first model, but it creates a representation bias. By leaving out smaller stations, our model ignores the demand in areas that might already have fewer services. In a real-world system, this could lead to an unfair distribution of bikes, where less popular neighborhoods get even less attention.

Furthermore, the data we use is completely anonymous and grouped by day and station. This means no personal user information is exposed. If we change our model in the future to use hourly data or specific user types (like registered members or casual users), we must keep this strict privacy standard.

In future steps of this project, we should also think about the social and economic background of the stations. We must

make sure that our model does not systematically disadvantage certain parts of the city.

IV. AI TOOL USE

AI tools were used for drafting support, restructuring text, and helping to standardize code and reporting outputs. The final content, decisions, code execution, result interpretation, and critical reflection were reviewed and organized by the authors.